

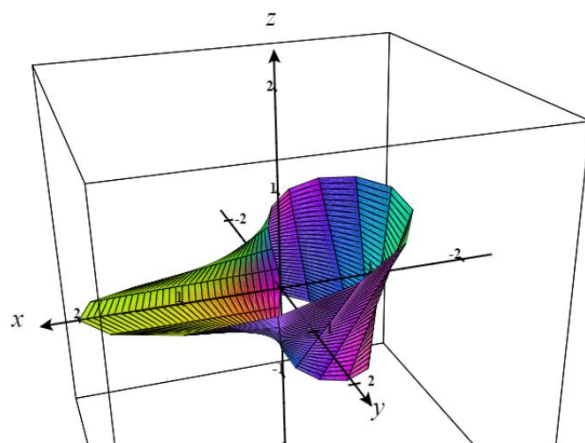
5.6 Applications of Surface Integrals; Flux

In this section we wish to introduce an important application of surface integrals in the field of physics. In order to do this, we first need to go over what it means for a surface to have two sides.

Demonstration: The instructor will show students an example of a Möbius strip. The key thing to take away from the demonstration is that the Möbius strip in some sense only has one side.

A parameterization of a Möbius strip is $\mathbf{r}(u, v) = \left\langle \left(1 + u \cos\left(\frac{v}{2}\right)\right) \cos(v), \left(1 + u \cos\left(\frac{v}{2}\right)\right) \sin(v), u \sin\left(\frac{v}{2}\right) \right\rangle$

where $-1 \leq u \leq 1$ and $0 \leq v < 2\pi$. A picture produced by a 3-d plotter is below.



Terminology: From this point forward in the course we consider only “two-sided” surfaces. We call a “two-sided” surface orientable. We now make this notion more precise.

Definition: Suppose that σ is a surface which has a tangent plane at every point (x, y, z) on σ except perhaps at the boundary. There are then two unit normal vectors at each such (x, y, z) on σ . If it is possible to choose a unit normal vector $\mathbf{n}$ at every such point (x, y, z) so that $\mathbf{n}$ varies continuously over σ , then σ is called orientable or an oriented surface. Choosing a particular $\mathbf{n}$ provides σ with an orientation.

Note: In order to have an oriented surface, $\mathbf{n}$ must vary continuously when it is in terms of x , y , and z ! If we have parameterized a surface in terms of u and v , it is possible that $\mathbf{n}$ varies continuously when it is in terms of u and v , but the surface is still not orientable.

Note: There is a nice intuitive demonstration as to why the Möbius strip is not orientable at:

http://mathinsight.org/moebius_strip_not_orientable

Note: Since $\mathbf{n}$ provides σ with an orientation, we will say “Consider σ with orientation determined by $\mathbf{n}$ ” when we are specifying which unit normal we are choosing.

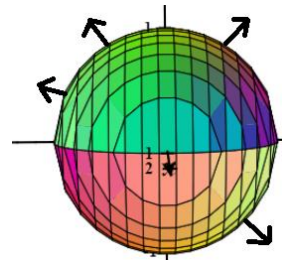
Facts:

1. There are two possible orientations for any orientable surface.
2. In section 4.4 (on tangent planes with parameterized surfaces) we discussed the idea that when a smooth surface σ is given by the parameterization $\mathbf{r}(u, v)$ the vector $\mathbf{r}_u \times \mathbf{r}_v$ would be normal to σ .
 - Therefore, if σ is orientable, an orientation of σ is given by $\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$ and the other possible orientation is determined by $-\mathbf{n}$.
3. In section 3.7 (on tangent planes) we discussed the ideas of creating vectors that were normal to given surfaces of the form $G(x, y, z) = k$. We saw that when a smooth surface σ is given by $z = g(x, y)$, we can let $G = z - g(x, y)$ and then the vector $\langle -g_x(x, y), -g_y(x, y), 1 \rangle$ is normal to the surface..
 - Therefore, if σ is orientable, an orientation of σ is given by $\mathbf{n} = \frac{\langle -g_x(x, y), -g_y(x, y), 1 \rangle}{\sqrt{(g_x(x, y))^2 + (g_y(x, y))^2 + 1}}$ and the other possible orientation is determined by $-\mathbf{n}$.

Note: For fact 2 the $\mathbf{n}$ determines upward orientation since the z component is positive.

Definition: A closed surface is a surface that is the boundary of a solid region E . For a closed surface, positive orientation is the one for which the normal vectors point outward from E .

A picture of positive orientation of a sphere is shown at the right.



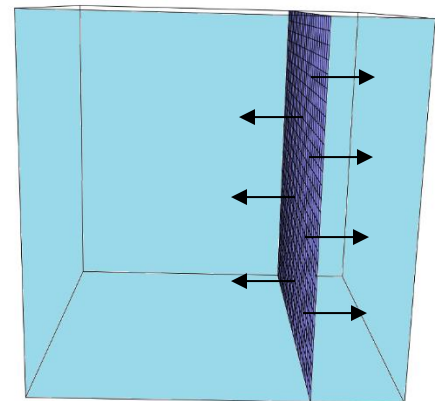
We are ready to introduce the important application of surface integrals encountered in physics.

Application:

Suppose that an oriented surface σ is immersed in an incompressible, steady-state fluid flow and that the surface is permeable so that the fluid can flow through it freely in either direction.

Would it be possible to find the net volume of the fluid that passes through the surface per unit of time? By *net volume* we mean the volume that passes through σ in the positive direction minus the volume that passes through σ in the negative direction.

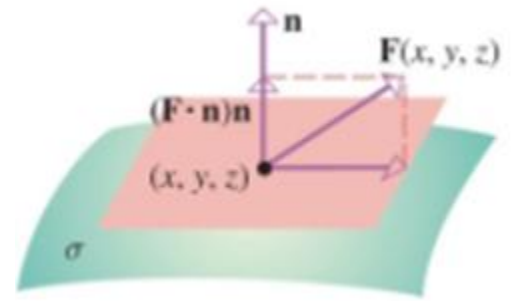
You can imagine scenarios of placing a sponge under water or dividing a tank of liquid with a permeable sheet.



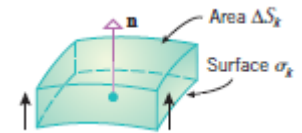
We want to make this idea more precise. Suppose that σ is a permeable surface that is immersed in a fluid and imagine that a fluid is flowing through σ at the point (x, y, z) at a rate determined by $\mathbf{F}(x, y, z)$.

Consider a small patch σ_k on the surface with surface area ΔS_k and choose a representative point (x_k^*, y_k^*, z_k^*) on this patch so that

- We can construct $\mathbf{n}_k$ which represents the unit normal vector on the positive side of σ which represents the direction in which the fluid flows.
- $\mathbf{F}(x_k^*, y_k^*, z_k^*) = \mathbf{F}_k$ can be used to represent the rate of the fluid flow for all points on this patch.
- $\mathbf{F}_k \cdot \mathbf{n}_k$ represents “how much” of the velocity of the liquid is in the direction of $\mathbf{n}_k$ and will contribute the fluid flow through the surface σ .
 - Since $\mathbf{F}$ may not be parallel to $\mathbf{n}$ we want to know “how much” of the velocity of the liquid is in the direction of $\mathbf{n}$ and will contribute the fluid flow through the surface σ versus “how much” of the velocity of the liquid is tangential to σ and thus will not contribute anything to the flow of the fluid through σ .
 - Based on our definition of $\mathbf{n}$ we know that:
 - if $\mathbf{F}_k \cdot \mathbf{n}_k > 0$ then the flow of the fluid is in the positive direction and
 - if $\mathbf{F}_k \cdot \mathbf{n}_k < 0$ then the flow of the fluid is in the negative direction.



We can then interpret $(\mathbf{F}_k \cdot \mathbf{n}_k) \Delta S_k$ to represent the amount of liquid that passes through σ per unit of time.



Therefore, the total volume passing through σ per unit of time is approximated as $V \approx \sum_{k=1}^n (\mathbf{F}_k \cdot \mathbf{n}_k) \Delta S_k$.

Thus we expect $V = \lim_{n \rightarrow \infty} \sum_{k=1}^n (\mathbf{F}_k \cdot \mathbf{n}_k) \Delta S_k = \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} dS$

The type of surface integral described in this application has a special name.

Definition: If $\mathbf{F}$ is a continuous vector field defined on an oriented surface σ with orientation $\mathbf{n}$, then the surface integral of $\mathbf{F}$ over σ is $\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S} = \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} dS$. The integral is also called the flux of $\mathbf{F}$ across σ .

Remark:

If we suppose that σ is a smooth surface with vector equation $\mathbf{r}(u, v)$ where $(u, v) \in D$, then we can calculate

$\mathbf{n}$ by computing $\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$ and then we can simplify the above result as follows:

$$\iint_{\sigma} \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F} \cdot \frac{(\mathbf{r}_u \times \mathbf{r}_v)}{\|\mathbf{r}_u \times \mathbf{r}_v\|} \|\mathbf{r}_u \times \mathbf{r}_v\| dA = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA$$

Note: One should see that the above application is quite similar to the application for finding the work done by a vector field on a particle, which was discussed in the section on line integrals.

Example 1: Let σ the sphere $x^2 + y^2 + z^2 = 1$ and $\mathbf{F}(x, y, z) = \langle 0, 0, z \rangle$. Find the flux of $\mathbf{F}$ across σ in the positive direction.

Using our knowledge from spherical coordinates that we can create a parameterization for σ .

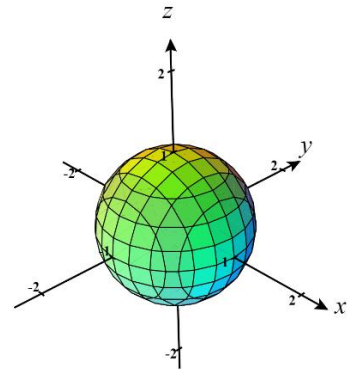
$$\mathbf{r}(u, v) = \langle \sin(u)\cos(v), \sin(u)\sin(v), \cos(u) \rangle \text{ where } 0 \leq u \leq \pi \text{ \& } 0 \leq v \leq 2\pi$$

Clearly,

$$\begin{aligned} \mathbf{r}_u \times \mathbf{r}_v &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos(u)\cos(v) & \cos(u)\sin(v) & -\sin(u) \\ -\sin(u)\sin(v) & \sin(u)\cos(v) & 0 \end{vmatrix} \\ &= \langle \sin^2(u)\cos(v), \sin^2(u)\sin(v), \sin(u)\cos(u) \rangle \end{aligned}$$

and so,

$$\begin{aligned} \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA \\ &= \int_0^{2\pi} \int_0^{\pi} \langle 0, 0, \cos(u) \rangle \cdot \langle \sin^2(u)\cos(v), \sin^2(u)\sin(v), \sin(u)\cos(u) \rangle \, du \, dv \\ &= \int_0^{2\pi} \int_0^{\pi} \sin(u)\cos^2(u) \, du \, dv \\ &= \int_0^{2\pi} \left(-\frac{1}{3}\cos^3(u) \right) \Big|_0^{\pi} \, dv \\ &= \int_0^{2\pi} \frac{2}{3} \, dv \\ &= \frac{4\pi}{3} \end{aligned}$$



Note: This example is identical to Example 2 from section 5.5. Note that $\mathbf{F} \cdot \mathbf{n}$ simplified to exactly what z^2 simplified to in Example 2 from section 5.5.

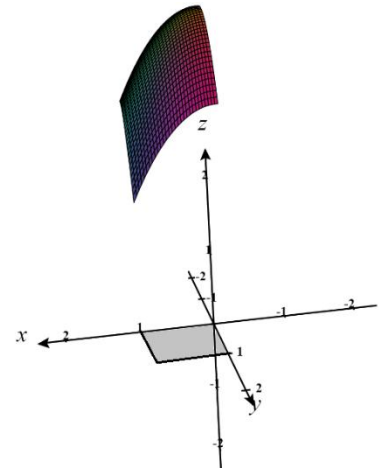
Example 2: Suppose that $\mathbf{F}(x, y, z) = \langle xy, yz, zx \rangle$. Find the flux of $\mathbf{F}$ across σ where σ is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square: $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$ with upward orientation.

We first create a parameterization for σ which is easily seen to be

$$\mathbf{r}(u, v) = \langle u, v, 4 - u^2 - v^2 \rangle \text{ where } D = \{(u, v) | 0 \leq u \leq 1, 0 \leq v \leq 1\}$$

Clearly, $\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -2u \\ 0 & 1 & -2v \end{vmatrix} = \langle 2u, 2v, 1 \rangle$ and so,

$$\begin{aligned} \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA \\ &= \int_0^1 \int_0^1 \langle uv, v(4 - u^2 - v^2), (4 - u^2 - v^2)u \rangle \cdot \langle 2u, 2v, 1 \rangle \, dudv \\ &= \int_0^1 \int_0^1 (2u^2v + 8v^2 - 2v^2u^2 - 2v^4 + 4u - u^3 - uv^2) \, dudv \\ &= \int_0^1 \left(\frac{2}{3}u^3v + 8v^2u - \frac{2}{3}v^2u^3 - 2v^4u + 2u^2 - \frac{1}{4}u^4 - \frac{1}{2}u^2v^2 \right) \Big|_0^1 dv \\ &= \int_0^1 \left(\frac{2}{3}v + 8v^2 - \frac{2}{3}v^2 - 2v^4 + 2 - \frac{1}{4} - \frac{1}{2}v^2 \right) dv \\ &= \left(\frac{1}{3}v^2 + \frac{8}{3}v^3 - \frac{2}{9}v^3 - \frac{2}{5}v^5 + \frac{7}{4}v - \frac{1}{6}v^3 \right) \Big|_0^1 \\ &= \frac{1}{3} + \frac{8}{3} - \frac{2}{9} - \frac{2}{5} + \frac{7}{4} - \frac{1}{6} \\ &= \frac{713}{180} \end{aligned}$$



Example 3: Find a formula for $\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S}$ if σ is an oriented surface with upward orientation given by $z = g(x, y)$ where $(x, y) \in D$ and $\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$.

We know that the orientation of σ is given by: $\mathbf{n} = \frac{1}{\sqrt{(g_x(x, y))^2 + (g_y(x, y))^2 + 1}} \langle -g_x(x, y), -g_y(x, y), 1 \rangle$.

So, using a formula from earlier, we have that:

$$\begin{aligned} \iint_{\sigma} \mathbf{F} \cdot d\mathbf{S} &= \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} dS \\ &= \iint_D \left(\mathbf{F}(x, y, z) \cdot \frac{\langle -g_x(x, y), -g_y(x, y), 1 \rangle}{\sqrt{(g_x(x, y))^2 + (g_y(x, y))^2 + 1}} \right) \sqrt{(g_x(x, y))^2 + (g_y(x, y))^2 + 1} dA \\ &= \iint_D (\mathbf{F}(x, y, z) \cdot \langle -g_x(x, y), -g_y(x, y), 1 \rangle) dA \end{aligned}$$

We notice that if we define a function $G(x, y, z) = z - g(x, y)$ then $\nabla G = \langle -g_x(x, y), -g_y(x, y), 1 \rangle$ and thus

$$\iint_{\sigma} \mathbf{F} \cdot \mathbf{n} dS = \iint_D (\mathbf{F} \cdot \nabla G) dA$$

Formula: If $\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$ is a continuous vector field defined on an oriented surface σ with upward orientation given by $z = g(x, y)$ where $(x, y) \in D$ and $G(x, y, z) = z - g(x, y)$, then the flux of $\mathbf{F}$ across σ is given by

$$\iint_D (\mathbf{F} \cdot \nabla G) dA$$

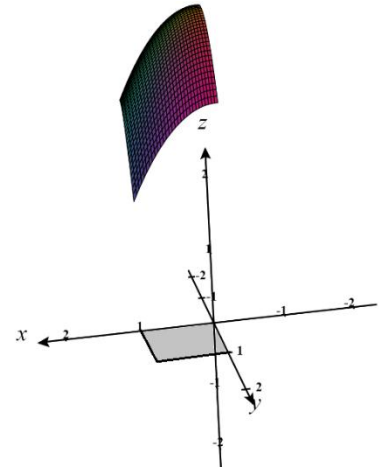
Question: What would the formula be if σ had downward orientation?

Note: In this section we have seen many formulas related to surfaces given by $z = g(x, y)$. It is very important to not view these formulas as new items that must be memorized. Rather, keep the general setting in mind. In particular, all the formulas come from the idea that such a surface can be parameterized as $\mathbf{r}(x, y) = \langle x, y, g(x, y) \rangle$. Then, one can use definitions to get at every formula.

Example 4: Suppose that $\mathbf{F}(x, y, z) = \langle xy, yz, zx \rangle$. Find the flux of $\mathbf{F}$ across σ where σ is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square: $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$ with upward orientation.

We use the previous formula and knowing that $G(x, y, z) = z - (4 - x^2 - y^2) = z - 4 + x^2 + y^2$ we have that:

$$\begin{aligned}
 \iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D \langle xy, yz, zx \rangle \cdot \langle 2x, 2y, 1 \rangle dA \\
 &= \int_0^1 \int_0^1 (xy(2x) + yz(2y) + zx) dx dy \\
 &= \int_0^1 \int_0^1 (2x^2y + 2y^2(4 - x^2 - y^2) + x(4 - x^2 - y^2)) dx dy \\
 &= \int_0^1 \int_0^1 (2x^2y + 8y^2 - 2x^2y^2 - 2y^4 + 4x - x^3 - xy^2) dx dy \\
 &= \int_0^1 \left(\frac{2}{3}x^3y + 8y^2x - \frac{2}{3}x^3y^2 - 2y^4x + 2x^2 - \frac{1}{4}x^4 - \frac{1}{2}x^2y^2 \right) \bigg|_{x=0}^{x=1} dy \\
 &= \int_0^1 \left(\frac{2}{3}y + 8y^2 - \frac{2}{3}y^2 - 2y^4 + 2 - \frac{1}{4} - \frac{1}{2}y^2 \right) dy \\
 &= \int_0^1 \left(-2y^4 + \frac{41}{6}y^2 + \frac{2}{3}y + \frac{7}{4} \right) dy = \left(-\frac{2}{5}y^5 + \frac{41}{18}y^3 + \frac{1}{3}y^2 + \frac{7}{4}y \right) \bigg|_{y=0}^{y=1} = \frac{713}{180}.
 \end{aligned}$$



Note: This answer is the same as the answer we found in Example 2!

Example 5: Let σ_1 be the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 4$ oriented by downward normal and $\mathbf{F}(x, y, z) = \mathbf{i} + y\mathbf{j} + \mathbf{k}$.

a) Compute the flux of $\mathbf{F}$ across σ_1 in the direction of orientation.

We first note that our surface looks like the picture to the right. Notice that this surface exists above the circle (shown in gray) of radius 2. Therefore, we are integrating over a region $D = \{(x, y) \mid x^2 + y^2 \leq 4\}$.

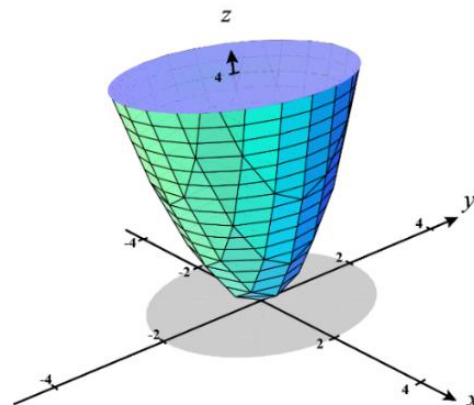
Since σ_1 is given by a function we let

$$G(x, y, z) = z - (x^2 + y^2) = z - x^2 - y^2 \text{ and note that}$$

Since $\nabla G = \langle -2x, -2y, 1 \rangle$ doesn't face in the downward direction (as indicated by the z -coordinate) we instead will choose to use $-\nabla G = \langle -2x, -2y, -1 \rangle$

We have that:

$$\begin{aligned} \iint_{\sigma_1} (\mathbf{F} \cdot \mathbf{n}) dS &= \iint_D (\mathbf{F} \cdot (-\nabla G)) dA \\ &= \iint_D \langle 1, y, 1 \rangle \cdot \langle -2x, -2y, -1 \rangle dA \\ &= \iint_D (2x + 2y^2 - 1) dA \end{aligned}$$



In order to more easily integrate over D we will convert to polar coordinates. Therefore, $D = \{(r, \theta) \mid 0 \leq r \leq 2 \text{ \& } 0 \leq \theta \leq 2\pi\}$ and we have:

$$\begin{aligned} \iint_D (2x + 2y^2 - 1) dA &= \int_0^{2\pi} \int_0^2 (2r \cos(\theta) + 2r^2 \sin^2(\theta) - 1) r dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (2r^2 \cos(\theta) + 2r^3 \sin^2(\theta) - r) dr d\theta \\ &= \int_0^{2\pi} \left(\frac{2}{3} r^3 \cos(\theta) + \frac{1}{2} r^4 \sin^2(\theta) - \frac{1}{2} r^2 \right) \bigg|_0^2 d\theta \\ &= \int_0^{2\pi} \left(\frac{16}{3} \cos(\theta) + 8 \sin^2(\theta) - 2 \right) d\theta \\ &= \int_0^{2\pi} \left(\frac{16}{3} \cos(\theta) + 4[1 - \cos(2\theta)] - 2 \right) d\theta \\ &= \left(\frac{16}{3} \sin(\theta) + 4\theta - 2 \sin(2\theta) - 2\theta \right) \bigg|_0^{2\pi} \\ &= (4\pi) - (0) \\ &= 4\pi \end{aligned}$$

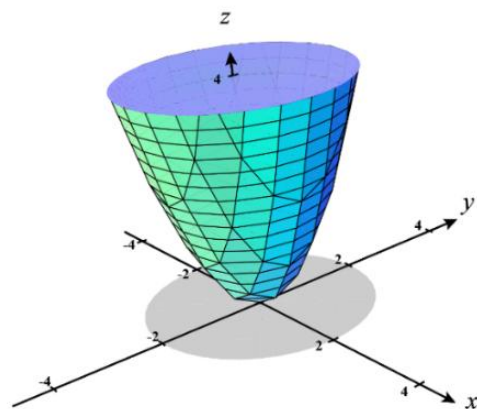
- b) Suppose that σ is the surface that bounds the region enclosed by $z = x^2 + y^2$ and $z = 4$ (essentially the surface σ_1 from part (a) in addition to the portion of $z = 4$ that encloses the solid region shown which we will call σ_2). Find the flux of $\mathbf{F}$ across σ with respect to outward normals.

Since flux integrals are just a special kind of surface integral, we know that we can simply compute the flux through each piece of the surface and then take the sum of these individual fluxes to find the total flux of $\mathbf{F}$ through σ .

$$\text{That is we know that } \iint_{\sigma} (\mathbf{F} \cdot \mathbf{n}) dS = \iint_{\sigma_1} (\mathbf{F} \cdot \mathbf{n}) dS + \iint_{\sigma_2} (\mathbf{F} \cdot \mathbf{n}) dS$$

$$\text{We already established in part (a) that } \iint_{\sigma_1} (\mathbf{F} \cdot \mathbf{n}) dS = 4\pi.$$

$$\text{Now we just need to compute } \iint_{\sigma_2} (\mathbf{F} \cdot \mathbf{n}) dS.$$



We note that since σ_2 is a portion of the surface $z = 4$ over the region $D = \{(r, \theta) | 0 \leq r \leq 2 \text{ \& } 0 \leq \theta \leq 2\pi\}$.

Since σ_2 is a function we let $G = z - 4$ and compute $\vec{\nabla} G = \langle 0, 0, 1 \rangle$.

We then obtain:

$$\begin{aligned} \iint_{\sigma_2} (\mathbf{F} \cdot \mathbf{n}) dS &= \iint_D (\mathbf{F} \cdot \vec{\nabla} G) dA = \iint_D \langle 1, y, 1 \rangle \cdot \langle 0, 0, 1 \rangle dA \\ &= \int_0^{2\pi} \int_0^2 r dr d\theta = \int_0^{2\pi} \left(\frac{1}{2} r^2 \right) \bigg|_0^2 d\theta = \int_0^{2\pi} 2 d\theta \\ &= 4\pi \end{aligned}$$

$$\text{Therefore, } \iint_{\sigma} (\mathbf{F} \cdot \mathbf{n}) dS = \iint_{\sigma_1} (\mathbf{F} \cdot \mathbf{n}) dS + \iint_{\sigma_2} (\mathbf{F} \cdot \mathbf{n}) dS = 4\pi + 4\pi = 8\pi$$

Note: The fact that both surface had the same flux here was coincidental and one should not expect the flux through all surfaces in a given scenario will produce the same result.

Note: Another application of the flux of $\mathbf{F}$ across σ would be in regards to electric flux.

Suppose that $\mathbf{E}(x, y, z)$ is an electric field where the units are Newtons per coulomb (N/C). Given a charge of q C located at (x, y, z) , $q\mathbf{E}(x, y, z)$ is the force acting on the charge.

The electric flux of $\mathbf{E}$ through the surface σ is $\iint_{\sigma} \mathbf{E} \cdot d\mathbf{S}$. Gauss's Law states that the net charge enclosed by a

closed surface σ is: $Q = \epsilon_0 \iint_{\sigma} \mathbf{E} \cdot d\mathbf{S}$ where ϵ_0 is called the permittivity of free space and

$$\epsilon_0 = 8.8542 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2).$$